TransUNetCD: A Hybrid Transformer Network for Change Detection in Optical Remote-Sensing Images

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***Abstract*— In the change detection (CD) task, the UNet architecture has achieved superior results. However, due to the inherent limitation of convolution operations, UNet is inadequate in learning global context and long-range spatial relations. Transformers can capture long-range feature dependencies, but the lack of low-level details may result in limited localization capabilities. Therefore, this article proposes an end-to-end encoding–decoding hybrid transformer model for CD, TransUNetCD, which has the advantages of both transformers and UNet. The model encodes the tokenized image patches from the convolutional neural network (CNN) feature map to extract rich global context information. The decoder upsamples the encoded features, connects them with higher-resolution multiscale features through skip connections to learn local–global semantic features, and restores the full spatial resolution of the feature map to achieve precise localization. The model proposed in this article not only solves the problem that redundant information is generated when extracting low-level features under the UNet framework, but also solves the problem that the relationship between each feature layer cannot be fully modeled and the optimal feature difference representation cannot be obtained. On this basis, we introduce a difference enhancement module to generate a difference feature map containing rich change information. By weighting each pixel and selectively aggregating features, the effectiveness of the network and the accuracy of extracting changing features are improved. The results on multiple datasets demonstrate that, compared to state-of-the-art methods, the TransUNetCD can further reduce false alarms and missed alarms, and the edge of the changing area is more accurate. The model has the highest score in each metric than other baseline models and has a robust generalization ability.**

***Index Terms*— Change detection (CD), optical remote-sensing image, transformer, UNet.**

I. Introduction

REMOTE-sensing image is the main data source of earth monitoring. Change detection (CD) is defined as the identification of multitemporal image differences in the same region [1]. Remote-sensing images are widely used in CD, including disaster assessment [2], [3], forestry and agricultural monitoring [4], [5], and urban planning [6], [7]. The development of CD technology makes it easier and faster to observe the world.

With the development of sensors, including satellite and airborne sensors, the temporal and spatial resolution is gradually improved, increasing the ability to acquire multitemporal remote-sensing images. Because high-resolution remote-sensing (HRSS) images can provide rich spatial information and fine surface details, it is of great significance to detect their changes for fine-grained applications such as precision agriculture and urban renewal. However, CD in HRSS images faces many challenges, such as pseudo-changes, scale change, and category imbalance. First, HRSS images have obvious geometric structure, complex texture information, and limited spectral information, resulting in spectral variation, spatial information loss, and bitemporal image geometric registration error. Therefore, there are a lot of pseudo-changes in the CD task, as shown in Fig. 1. Affected by seasonal changes and illumination conditions, complex buildings have obvious nonuniform spectral characteristics. In Fig. 1(a), the roof color varies greatly, which may cause uncertainty in the interpretation process. The presence of tree shadow in Fig. 1(a) and building shadow in Fig. 1(c) may obscure some ground objects and cause information loss, and the difference in shadow shape may be misjudged as change. In Fig. 1(b), the presence of temporary objects such as cars also affects the CD results. Due to different flight angles of sensors, some high-rise buildings in Fig. 1(c) have obvious spatial displacement, resulting in geometric registration difficulties. In addition, datasets of real remote-sensing images have different scales, so how to adapt to different scales to maintain the robustness and generalization ability of the algorithm needs attention. Finally, in the CD task, the number of unchanged pixels is far more than that of changed pixels, which can be regarded as a typical unbalanced problem. Therefore, how to effectively extract and learn rich feature information, reduce the interference of pseudo-changes, and improve the detection accuracy are important problems for HRSS image CD at present.

According to the detection unit, traditional CD can be divided into pixel- and object-based methods. Pixel-based methods directly compare the pixel values of multitemporal images to generate difference maps and then

Manuscript received December 30, 2021; revised March 8, 2022 and April 4, 2022; accepted April 19, 2022. Date of publication April 22, 2022; date of current version May 9, 2022. This work was supported by the National Natural Science Foundation of China for the Project Automatic Acquisition and Conformal-Mapping Processing of 3-D Laser Point Cloud of Immovable Cultural Relics under Grant 42071444. *(Corresponding author: Ruofei Zhong.)*

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Digital Object Identifier 10.1109/TGRS.2022.3169479

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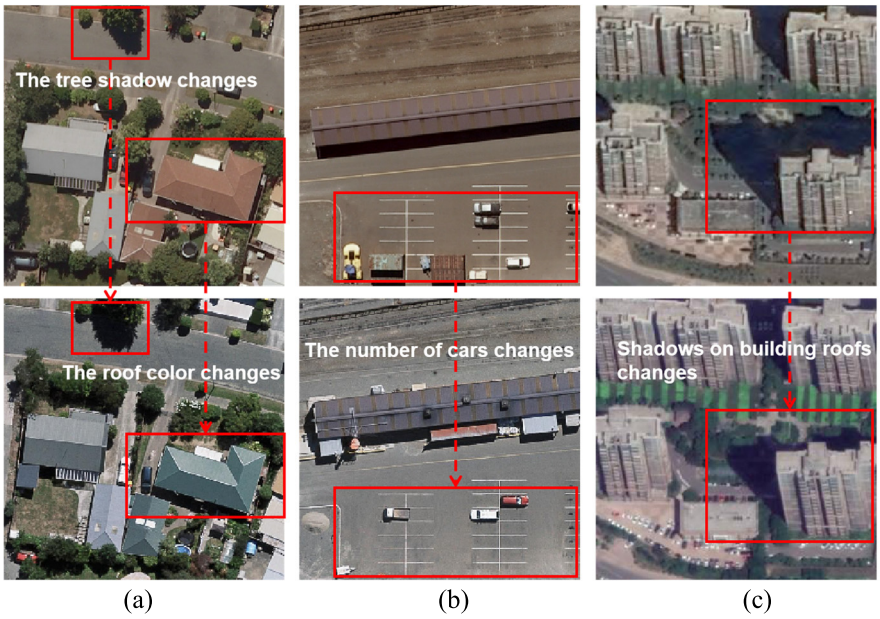


Fig. 1. Variety of pseudo-changes affects CD results. (a) Tree shadow changes and the roof color changes. (b) Moving cars changes. (c) Shadows on high-rise building roofs changes.

generate change maps according to their thresholds. It mainly includes arithmetic-based (such as image difference [8], image rationing), transformation-based (such as change vector analysis (CVA) [9], [10], principal component analysis (PCA) [11], [12], independent component analysis (ICA) [13]), image classification (post-classification methods), and other algorithms (slow feature analysis (SFA) [14], multivariate alteration detection (MAD) [15]). Pixel-based methods focus on the spectral change of a single pixel, ignore the spatial context information, and face some problems such as determining the empirical threshold. Object-based methods segment the image according to the shape, spectrum, and texture and compare the classification result to get the change map. Object-based methods consider the spatial and spectral information of the original image comprehensively. Still, the misclassification error caused by the uncertainty of the segmentation object propagates in different stages of CD, which may limit the detection accuracy. In addition, pixel- and object-based methods require a lot of artificial interference and are easily affected by sensors and illumination conditions.

With the rise of deep learning technology, many new CD algorithms have emerged. Convolutional neural networks (CNNs) have been successfully applied to CD algorithms with strong discriminant ability. The method based on CNNs converts bitemporal images into high-level spatial deep features and extracts the highly abstract semantic context of the changing area, which minimizes error propagation and reduces artificial errors caused by preprocessing. According to the fusion strategy, it can be divided into early fusion (EF) [16]–[20] and late fusion [21]–[28] networks. In the EF network, the registered multitemporal images are input into a deep network as a whole. The EF network can capture the hierarchical difference representation, but it lacks the deep feature of individual original images, resulting in the destruction of the boundary of the change map. Different from the EF network, the late fusion network (Siamese network) consists of two networks with shared weights, which learns single-temporal features, respectively, and then connects them as the input of the CD layer. But the presence of gradient disappearance prevents the gradient from flowing backward, making it impossible for the shallower layers to learn useful features. This network structure may lead to insufficient learning and low representativeness of original image features. Therefore, how to combine the advantages of early and late fusion networks and fully learn the deep information of low and high levels is another key problem to be solved.

In order to solve the above problems and obtain better deep features, it has become a trend to design deeper and more complex CNN networks, but at the same time, it brings about explosive growth parameters. Therefore, many scholars proposed various improved models to obtain more discriminant feature representation, such as expanding the receptive field, combining the generative adversarial network (GAN) [29]–[31] model or the recurrent neural network (RNN) [32], [33] model, and adopting the spatial or channel attention mechanism [17], [22], [26]–[28]. These methods improve the precision of CD, but because of the inherent limitations of the convolution, the receptive field of CNNs is limited to the image pixels to which it is linked. They are still insufficient in long-range concepts in space–time dimension. While focusing on the extracted semantic features of deep changes, CNNs ignore the importance of shallow information with high resolution and the characteristics of fine particles, resulting in the uncertainty of edge pixels of changing areas and the loss of small objects. Self-attention [34] shows the ability of nonlocal operation of images in spatial–temporal dimension, and long-range dependencies captured can obtain more distinguishable feature representation.

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Therefore, this article proposes TransUNetCD, an end-to-end encoding–decoding CD model combining transformers and UNet. Based on the UNet network architecture, the transformer encodes the tokenized image patches from the CNN feature map to model the context and extract rich global context information. The decoder upsamples the encoded features, connects them with the high-resolution multiscale features through skip connections to learn local–global semantic features, restores the full spatial resolution of the feature map, and achieves accurate localization. Based on this, a difference enhancement module (DEM) is introduced to enhance the change information. In addition, a new loss function is proposed to mitigate the effects of class imbalance. TransUNetCD can not only obtain rich context correlation, but also retain important information of shallow features better, which improves the accuracy of the model and relieves the influence of pseudo-changes. To sum up, the main contributions of this article are as follows.

1) We propose an end-to-end network architecture combining the transformer and UNet, which is the first attempt at combining the transformer and UNet in the CD field. The proposed model combines the advantages of early and late fusion strategies through the UNet structure and the Siamese CNN network and achieves full learning of shallow and high deep features.
2) We introduce a transformer module to fully learn the global context and extract more representative deep features, which reduces the interference of pseudo-changes. We propose a DEM to obtain a more discriminative feature representation by weighting the change map, which can generate the difference change map containing rich change information and improves the learning ability of the model. We propose a weighted loss function to solve the sample imbalance problem.
3) Experimental results show that the proposed model can better locate detailed information and texture features, return change areas with complete boundaries, and achieve better information retention. In the comparison of different datasets results, the proposed model is significantly superior to the comparison model.
4) The proposed model performs well in the CD of high- and medium-resolution remote-sensing images. The model has an excellent performance in the CD of ground objects such as buildings and roads and can well weaken the influence of seasonal change. It is mainly used in urban planning, disaster monitoring, and other application fields.

The rest of the sections are structured in this article as follows. Section II reviews the related work. Section III introduces the proposed TransUNetCD method. Section IV presents experiments and results analysis. Finally, Section V presents the conclusion of this article.

II. Related Work

A. Visual Transformer

Transformers were first applied in natural language processing (NLP) [35] and are deep neural networks based on the self-attention mechanism. They are now receiving more and more attention in image processing due to their powerful representation ability. Unlike CNNs, which rely on local operators to extract information, transformers use stacked multihead attention blocks to model global relationships between tokenized image patches. This allows them to drill down into the long-range dependencies that the data might present [36]. Transformers have been used in image classification [34], object detection [37], semantic segmentation [38], [39], image recognition [40], and video captioning [41], [42].

With the gradual development of transformers, the research has also been carried out in the field of remote sensing, mainly focusing on image classification [43]–[45], object detection [46], semantic segmentation [47], [48], and image fusion [49]. In terms of image classification, Li *et al.* [44] proposed a CNN-transformer approach to perform the crop classification and verified the results in multisensor Sentinel-2A/B and Landsat-8 images. Deng *et al.* [45] proposed a high-performance joint framework that combined CNNs and vision transformer (ViT) to further boost the discriminative ability of features for HRSS scene classification. In terms of semantic segmentation, Ding *et al.* [47] proposed a wide-context network (WiCoNet) for semantic segmentation in HRSS images, which introduced a context transformer to capture long-range context information. Wang *et al.* [48] proposed a semantic segmentation scheme based on the transformer for fine-resolution remote-sensing images, which introduced Swin Transformer as the backbone to extract the context information. In terms of image fusion, Li *et al.* [49] proposed a multistream fusion network for remote-sensing spatial–temporal fusion based on the transformer and convolution.

In the field of CD, there are also transformer-related articles. Chen *et al.* proposed a bitemporal image transformer (BiT) method to model contexts within the spatial–temporal domain. However, this method ignores the importance of high-resolution shallow information and directly obtains the full resolution change map through upsampling, which results in the loss of low-level details. Our model uses a cascade upsampling decoder to obtain full-resolution results, which can better locate the details and texture features, so the boundaries of changing areas are more complete.

B. CD Method Based on Deep Learning

CNNs perform well in feature extraction, so it is widely used in CD algorithms, such as SegNet [50] and UNet [16]–[18]. Lei *et al.* [51] proposed a symmetric UNet network for landslide mapping, in which they used a pyramid pooling module to obtain multiscale change information. Peng *et al.* [16] proposed a UNet++ network with the fusion strategy of multiple side outputs. Daudt *et al.* [21] presented three fully convolutional (FC) networks based on the UNet model, including FC-EF, FC-Siam-conc, and FC-Siam-diff. FC-EF adopted the EF strategy, while FC-Siam-conc and FC-Siam-diff adopted the late fusion strategy, in which FC-Siam-conc fused the features via

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concatenation, while FC-Siam-diff fused the features via difference.

Although these methods have demonstrated their ability in CD, the existing methods lack global feature extraction. These methods pay little attention to the spatial context information and the internal relationship between high- and low-level features. The features obtained are usually sensitive to factors such as noise, angle, shadow, and context and thus are less robust to pseudo-changes. Therefore, many improved algorithms are proposed to better encode image context in time–space dimension and improve feature discrimination ability, including stacking more convolution layers [52], using dilated convolutions [53], combining GAN models [29]–[31], using RNNs [32], [33], and using the attention mechanism [17], [22], [26]–[28].

RNNs can fully mine change information and generate spatial–temporal spectral features with rich change information. Papadomanolaki *et al.* [32] integrated long- and short-term memory blocks (LSTMs) into a FC network to detect urban changes for Sentinel-2 images, proving that the RNN is effective for capturing spectral or temporal relationships between images. Khusni *et al.* [33] proposed a model combining UNet and RNNs for time modeling. However, the RNN has some limitations in CD due to infrequent spectral information and insufficient time information of high-resolution images.

The attention mechanism can automatically weight the feature map, enhance the changed features, and weaken the unchanged features, including spatial attention (SA), channel attention, positional attention, and self-attention [18], [54]. Peng *et al.* [17] proposed the difference-enhancement dense-attention convolutional neural network (DDCNN), which captured the features of object changes by introducing spatial and channel attention. Zhang *et al.* [22] proposed a CD network [image fusion network (IFN)] based on deep supervision and used the attention module for feature enhancement. Currently, attention-based methods only enhance features in channel or spatial scales, ignoring shallow feature enhancement, which is insufficient to learn global context and has high computational complexity. In this article, we apply the transformer module to obtain dense context information. The transformer is an extension of self-attention, adding weight tying, feed-forward networks (FFNs), and positional encoding. We use weight tying to significantly reduce computational overhead in the model. FFNs provide a larger dimension of feature space for the transformer and enhance feature expression for the tokenized image patches. Positional encoding is used to provide the relative spatial position of the image. Therefore, compared with using only self-attention, the feature representation of changing information can be better with less computation. In addition, compared with the soft attention mechanisms such as channel attention and SA, self-attention uses the intrinsic information of features as much as possible for attentional interaction and reduces the dependence on external information. Therefore, self-attention is better at capturing the internal correlation of features, which can solve the problem of long-term dependence that soft attention cannot solve.

III. Proposed Method

In this section, we first introduce the proposed network architecture, followed by the main network architecture transformer encoder, the cascading upsampling decoder (CUD), and the proposed DEM. Finally, a detailed description of model training, including the training process and loss function, is introduced.

A. Network Architecture

CD can be regarded as an extension of image segmentation. The network architecture based on UNet performs well in experiments. UNet combines high-level semantic information with low-level texture information through skip connections to achieve feature extraction and detail recovery. However, although CNNs have achieved excellent performance, global long-range semantic information cannot be well learned because of the locality of convolution operations. The transformer uses the self-attention mechanism. The self-attention mechanism provides an efficient way to capture global context information through triples (key, query, value), so the transformer has efficient spatial–temporal modeling capability of global semantic relationships, which is conducive to feature representation of interest changes. Inspired by the above, this article introduces the transformer into the design of the UNet encoder to realize the CD of optical remote-sensing images. The overall flow of the TransUNetCD model is shown in Fig. 2. The proposed network first extracts the deep features of bitemporal images as an independent stream using several convolution units. We concatenate the bitemporal deep features and split the features into a sequence of patches. Each patch is flattened into a 1-D vector and then linearly projected to a patch embedding to produce a sequence of patch embeddings. Then we encode patch embeddings with the transformer. The network learns and correlates the global context of high-level semantic concepts in the transformer encoder to obtain the enhanced deep features. Next, we carry out the CUD to achieve accurate positioning. That is, the shallower layers containing the local structure information of bitemporal images are concatenated to the deeper layers of the same scale by skip connections. Finally, we use the dot product operation between the obtained feature map and the difference image obtained by the DEM to obtain the change map with fine image details.

B. CNN-Transformer Encoder

Given bitemporal remote-sensing images $\mathbf{x_1}, \mathbf{x_2} \in \mathbb{R}^{H\times W\times C}$ with a spatial resolution of $H \times W$ and channel number of C. TransUNetCD first uses CNNs as a feature extractor to extract deep features $\mathbf{y_1}, \mathbf{y_2} \in \mathbb{R}^{H'\times W'\times C'}$ of $\mathbf{x_1}$, $\mathbf{x_2}$ with a spatial resolution of $H' \times W'$ and channel number of C'. We concatenate $\mathbf{y_1}$ and $\mathbf{y_2}$ to obtain deep features $\mathbf{y} \in \mathbb{R}^{H'\times W'\times 2C'}$, which can provide feature mapping for the transformer. The experiment proves that the concatenation operation can improve the CD accuracy more than the difference operation. See fusion experiment for details. Then patch embedding is applied to the patches extracted from the CNN feature map instead of the raw image.

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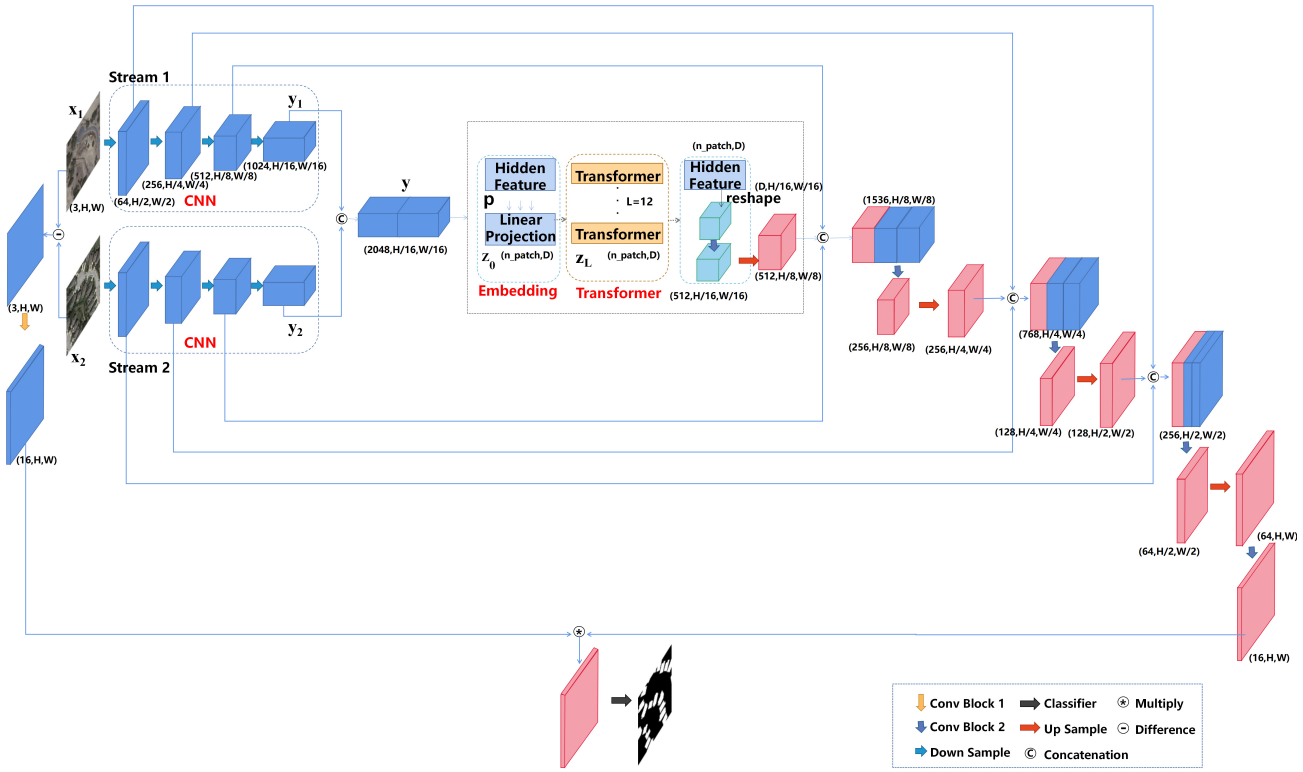


Fig. 2. TransUNetCD.

We start with sequentialization. The input $\mathbf{y}$ is reshaped into a flat 2-D patch sequence $\mathbf{p} = [p_1 \cdots p_N] \in \mathbb{R}^{N \times P^2 \times 2C'}$ for performing tokenization, where the patch size is $P \times P$ and the number of patches is $N = HW/P^2$.

Then we map the vectorized patches $\mathbf{p}$ to the latent D-dimensional embedding space by using a trainable linear projection. In order to encode the patch spatial positional information, we learn the specific position embedding and add it to the patch embedding to retain the position information. The specific steps are as follows:

$$\mathbf{z_0} = \left[p_1\mathbf{E_1}; p_2\mathbf{E_2}; \cdots; p_N\,\mathbf{E_N}\right] + \mathbf{E}_{\text{pos}} \tag{1}$$

where $\mathbf{E} = [\mathbf{E}_1 \cdots \mathbf{E}_N] \in \mathbb{R}^{(P^2 \cdot 2C') \times D}$ represents patch embedding projection and $\mathbf{E_{pos}} \in \mathbb{R}^{N \times D}$ represents position embedding.

After obtaining the patches of deep features $\mathbf{z_0}$, we use the transformer encoder to conduct context modeling. The transformer has efficient spatial–temporal modeling ability of global semantic relations and can generate context-rich feature representation.

We design the transformer model similar to the Vision Transformer (ViT) and slightly different from the NLP Transformer, as shown in Fig. 3. We put layer normalization before multihead self-attention (MSA) and multilayer perceptron (MLP) to make sure the input values are not too big to process.

The transformer encoder composed of L layers is applied to the sequence of tokens $\mathbf{z_0}$ to generate a sequence of contextualized encodings $\mathbf{z_L}$. A transformer layer consists of an MSA block and an MLP block. The MSA module is used to

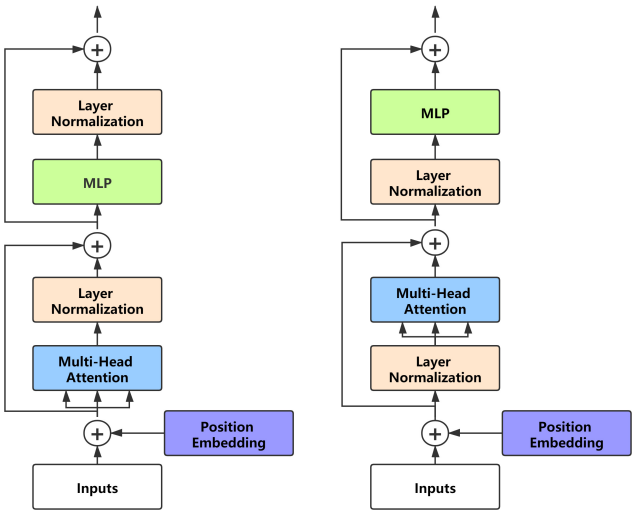


Fig. 3. (Left) NLP Transformer. (Right) Vision Transformer.

provide an effective modeling method to capture global context information. Multihead allows the model to jointly attend to information from different representation subspaces at different positions. The MLP module adds nonlinear transformation capability to the transformer. By activating functions, the transformer enhances the ability to express features

$$\begin{aligned} \mathbf{z}_{i'} &= \text{MSA}\left(\text{LN}\left(\mathbf{z}_{i-1}\right)\right) + \mathbf{z}_{i-1} \\ \mathbf{z}_i &= \text{MLP}\left(\text{LN}\left(\mathbf{z}_{i'}\right)\right) + \mathbf{z}_{i'} \end{aligned} \quad i \in \{1, \ldots, L\} \tag{2}$$

where $\text{LN}(\cdot)$ denotes the layer normalization operator and $\mathbf{z}_i$ is the encoded image representation. Let us start with single-head

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self-attention. Self-attention maps the original feature map to three vectors, namely query (Q), key (K), and value (V). First, we calculate the correlation weight matrix coefficients of Q and K. Second, the weight matrix is normalized by soft operation. Finally, the weight coefficients are superimposed on V to realize the modeling of global context information. Self-attention is computed as follows:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \tag{3}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices, with size $N \times d$, where N and d denote the number of patches and the dimension of single head attention.

Instead of simply applying a single self-attention function, the transformer uses MSA. By separately calculating each head attention, the results of multihead attention are finally concatenated together as the final output of the MSA module:

$$\begin{aligned}\text{MSA}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{Concat}(\text{head}_1,\ldots,\text{head}_H)\mathbf{W}^{\mathbf{O}}\\ \text{where}\quad \text{head}_i &= \text{Attention}\left(\mathbf{Q}\mathbf{W}_i^{\mathbf{Q}},\mathbf{K}\mathbf{W}_i^{\mathbf{K}},\mathbf{V}\mathbf{W}_i^{\mathbf{V}}\right)\end{aligned} \tag{4}$$

where $\mathbf{W}_i^{\mathbf{Q}} \in \mathbb{R}^{D\times d_k}$, $\mathbf{W}_i^{\mathbf{K}} \in \mathbb{R}^{D\times d_k}$, $\mathbf{W}_i^{\mathbf{V}} \in \mathbb{R}^{D\times d_v}$, and $\mathbf{W}^{\mathbf{O}} \in \mathbb{R}^{hd_v\times D}$ are parameter matrices. Here, h and D represent the number of attention heads and the dimension of embedding layers, respectively, and for each of these, we use $d_k = d_v = D/h$.

C. Cascading Upsampling Decoder

In general, a simple upsampling of encoder features to full resolution can yield good detection results [23], but this is not the best use of the transformer for the CD because $H/P \times W/P$ is usually much smaller than the original image resolution $H \times W$, thus inevitably leading to the loss of low-level details. Therefore, in order to compensate for information loss, we use a CUD for precise positioning.

The CUD decodes the encoded features by several cascading upsampling blocks. First, we reshape the sequence of patch encodings $\mathbf{z}^{\mathbf{L}} \in \mathbb{R}^{HW/P^2\times D}$ into $H/P \times W/P \times D$. The spatial resolution $H \times W$ is then restored using cascading upsampling blocks. Like the UNet, the CUD uses skip connections to connect the multiscale features of the encoder with the upsampling features. We concatenate shallow features and deep features together to reduce the loss of spatial information caused by down-sampling.

Each block consists of a $2\times$ upsampling operator, a conv block, which consists of two 3×3 convolution layers, a batch normalization (BN) layer, and a ReLU layer successively. The upsampling method uses bilinear interpolation, and the dimension of the concatenation feature of the convolution layer is consistent with that of the upsampling feature, as shown in Fig. 4.

Feature1 and feature2 in Fig. 2 represent the output of each convolution layer of stream 1 and 2, respectively.

D. Difference Enhancement Module

The difference map of bitemporal images can clearly show the image changes, but it uses undifferentiated pixel calculation and ignores the difference in the local feature of images.

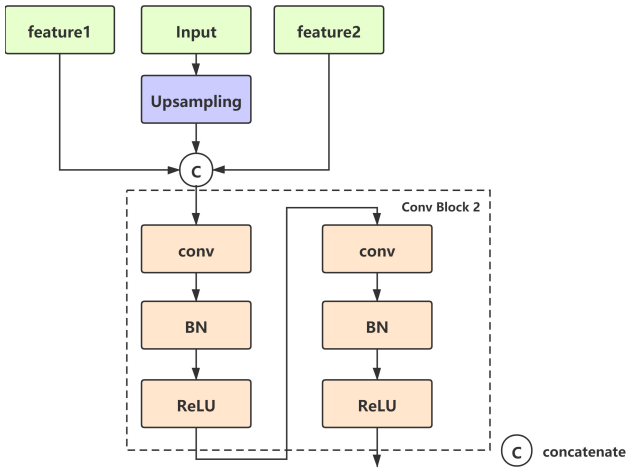


Fig. 4. CUD.

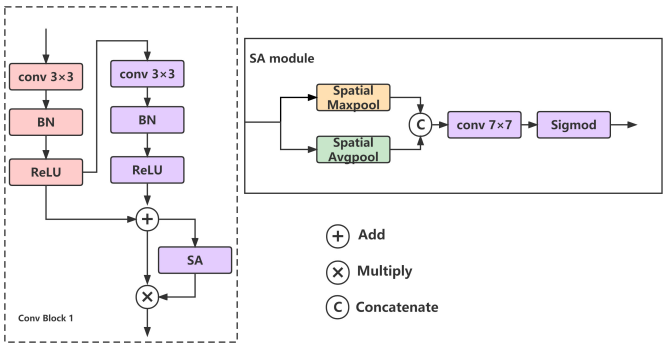


Fig. 5. DEM.

The presence of radiation and geographical errors affects its accuracy, so it cannot be used directly. The difference map can measure the difference between the input bitemporal images and improve the learning ability of the network. Therefore, we introduce a DEM to provide weight for deep features, consisting of two convolution units and an SA module. The SA module can distinguish changing objects and backgrounds in deep features by learning the SA map of difference images, as shown in Fig. 5.

The SA module uses average pooling and max pooling of the features after convolution operations and generates two matrices of size $1\times H \times W$. Then we use concatenation operations and a 7×7 convolution operation. Finally, a sigmoid function is used to obtain the final SA map and the equation is as follows:

$$M_s(F) = \sigma\left(f^{(7\times7)}(\text{AvgPool}(F);\ \text{MaxPool}(F))\right) \tag{5}$$

where F represents the features after convolution units.

E. Network Details

1) Training Process: The training process includes encoding and decoding. The encoder includes two stages: a CNN and a transformer. In the CNN stage, we use the improved ResNet-50 as the backbone network for feature extraction of bitemporal images. Different from the original ResNet-50, our

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ResNet has changed from five to four stages. It contains the first stage of preprocessing and the last three stages with three, four, and nine bottlenecks. After the CNN operation, the channels of the feature map are 64, 256, 512, and 1024. The dimension ($W \times H \times C$) of the feature changed from $256 \times 256 \times 3$ to $16 \times 16 \times 1024$. In sequentialization and patch embedding stages, the setting is similar to ViT with a hidden layer size of 768 and patch size of 16. In the transformer process, we set the number of transformer layers to 12, the MLP scale to 3072, and the activation function to gelu.

In the decoding process, we use concatenation to make skip connections with the features of the same dimension, and we set the number of skip connections as 3. We need to cascade four $2\times$ upsampling blocks consecutively in the CUD to reach the full resolution. The channel of the output feature map in the four decoding stages are 256, 128, 64, and 16.

2) Loss Function: In CD tasks, there is a large bias in the distribution of changed and unchanged pixels, which leads to a serious sample imbalance problem. In order to improve the detection performance and reduce the impact of sample imbalance, we introduce dice coefficient loss (DICE) based on the weighted binary cross-entropy (WCE) loss function

$$L = L_{\text{wce}} + \lambda L_{\text{dice}} \tag{6}$$

where L_{wce} denotes weighted binary cross-entropy loss, L_{dice} denotes dice coefficient loss, and λ refers to the weight that balances the two losses.

Binary cross-entropy loss is commonly used for binary classification problems. To solve the problem of sample imbalance, WCE adds weight to the changed and unchanged classes, defined as

$$L_{\text{wce}} = -t_i w_1 \log(y_i)\text{-}(1 - t_i) w_2 \log(1 - y_i) \tag{7}$$

where t_i represents the ground truth of pixel i, and $t_i = 1$ and $t_i = 0$ represent changed and unchanged categories, respectively. y_i represents the predicted probabilities of changed categories, and $1 - y_i$ represents the predicted probabilities of unchanged categories. w_1 represents the weight of changed categories, and w_2 represents the weight of unchanged categories. Here, we use pixel proportions to represent weights.

In image segmentation tasks, a loss function based on the region is proposed to solve the problem of sample imbalance, such as dice loss and IoU loss. In the CD task, dice loss results are relatively good. The dice coefficient is defined as the similarity of two contour regions, defined as

$$L_{\text{dice}} = 1 - \frac{2 y_i t_i}{y_i + t_i} \tag{8}$$

where y_i and t_i represent the prediction probability belongs to the changed category and ground-truth value, respectively.

IV. Experiments

A. Datasets and Experimental Setup

1) Datasets: We select six different CD datasets for algorithm verification and verify the robustness of the algorithm under different resolutions, different scenes, seasonal changes, spectral variability, and information loss.

a) WHU building change detection dataset (WHU-CD) [6]: The WHU-CD dataset is a building dataset of aerial images, which consists of one pair of high-resolution (0.2 m) aerial images with $32\,507 \times 15\,354$ pixels. The dataset covers Christchurch, New Zealand, and only includes buildings, excluding large cars, containers, greenhouses, and other man-made structures. We divide the data into 256×256 pixel patches without overlapping and randomly divide it into 5949 training sets, 742 validation sets, and 743 test sets, respectively.

b) Synthetic images and real season-varying change detection dataset (CDD) [55]: The CDD dataset is the season-varying dataset of Google Earth images, which consists of seven pairs of season-varying images with 4725×2700 pixels and four pairs of season-varying images with the size of 1900×1000 pixels. The spatial resolution of the dataset is $0.03 \sim 1$ m/pixel. This dataset contains the changes in different objects (such as buildings, roads, and vehicles). It ignores the changes caused by seasonal differences and brightness, which poses a great challenge to the CD algorithm. The dataset size is 256×256 pixels and contains 10 000 training sets, 3000 validation sets, and 3000 test sets, respectively.

c) LEVIR building change detection dataset (LEVIR-CD) [56]: The LEVIR-CD dataset is a building dataset of Google Earth images, which consists of 637 pairs of high-resolution (0.5 m) images with 1024×1024 pixels. This dataset has significant land-use changes from 20 different regions that sit in several cities in Texas of the US, especially the building growth (the change from soil/grass/hardened ground or building under construction to new build-up regions) and the building decline. It covers various types of buildings, such as villa residences, tall apartments, small garages, and large warehouses. We divide the data into 256×256 pixels patches and obtain 3169 training sets, 438 validation sets, and 937 test sets, respectively.

d) DSIFN change detection dataset (DSIFN) [22]: The DSIFN dataset is a season-varying dataset of Google Earth images, which consists of six large coregistered bitemporal image pairs that are collected from six major cities in China. The spatial resolution of the dataset is 2 m/pixel. This dataset includes changes such as roads, buildings, farmland, and water. It contains different brightness and contrast images and different land objects from six cities, which makes it difficult to detect changes. We divide the data into 256×256 pixel patches and obtain 14 400 training sets, 1360 validation sets, and 192 test sets, respectively.

e) Onera satellite change detection dataset (OSCD) [57]: The OSCD dataset is a multispectral dataset of Sentinel-2 satellites images from different countries, which consists of 24 pairs of multispectral images between 2015 and 2018. The spatial resolution of the dataset varies between 10, 20, and 60 m. The changes focus on urban changes, such as new buildings or roads. We divide the data into 256×256 pixel patches with overlapping the adjacent image patches by 50% and process the data with rotation and image blurring.

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We obtain 3524 training sets, 408 validation sets, and 406 test sets, respectively.

f) Satellite side-looking dataset (S2Looking) [59]: The S2Looking dataset is a satellite side-looking dataset of the GaoFen (GF), SuperView (SV), and BeiJing-2 (BJ-2) satellites from 2017 to 2020, which consists of 5000 pairs of high-resolution (0.5–0.8 m) images with 1024×1024 pixels. This dataset presents a unique, large-scale, side-looking, remote-sensing dataset for building CD covering rural areas around the world. It significantly expands upon the degree of richness offered by available datasets, by providing: 1) larger viewing angles; 2) large illumination variances; and 3) the added complexity of the characteristics encountered in rural areas. We divide the data into 256×256 pixel patches and obtain 56 000 training sets, 8000 validation sets, and 16 000 test sets, respectively.

2) Parameter Setting: Our model is implemented under the Pytorch framework, using a single NVIDIA GeForce GTX 1080Ti GPU with 64 G memory for training and testing. We use the stochastic gradient descent (SGD) method with an initial learning rate of 0.01, the momentum of 0.9, and the weight decay of 5e-4 for model optimization. We adopt the linear learning rate decay scheduler. Moreover, the batch size of our model is set to 12, and the other comparison model is set to 4. The same training and testing settings are adopted on all datasets.

3) Evaluation Metrics: We use precision (P), recall (R), Intersection over Union (IoU), and F1-score to quantitatively evaluate the performance of our model. The above metrics are defined as follows:

$$\begin{aligned} P &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ R &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ \mathrm{IoU} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \\ \text{F1-score} &= \frac{2\mathrm{PR}}{P+R} \end{aligned} \tag{9}$$

where true positive (TP), false positive (FP), false negative (FN), and true negative (TN) represent the number of unchanged pixels detected correctly, unchanged pixels unpredicted, changed pixels unpredicted, and changed pixels detected correctly, respectively. Among them, F1-score and IoU reflect the generalization ability of the model better.

4) Comparative Method: In order to verify the effectiveness of our model, we compare some state-of-the-art methods, such as U-net++_multiple side-output fusion (MSOF) [16], IFN [22], DDCNN [17], and BiT [23].

a) U-net++_MSOF: A UNet++ network with an EF strategy, which employs dense skip connections to learn multilevel features. Meanwhile, this network uses MSOF to extract the spatial details better. We set this model as the benchmark method.

b) IFN: An attention-based network with a late fusion strategy, which employs a FC two-stream architecture to extract deep features and uses a deeply supervised difference discrimination network (DDN) to identify changes. Meanwhile, this network adds the channel and SA modules to adaptively emphasize important features while suppressing irrelevant features.

c) DDCNN: An attention-based network with an EF strategy, which integrates a dense attention method consisting of several up-sampling attention units to obtain more discriminative features.

d) BiT: A transformer-based network with a late fusion strategy, which adds the transformer to better model contexts.

B. Results Analysis and Benchmark Comparison

1) Accuracy Comparison: We use TransUNetCD to detect changes in four datasets of WHU-CD, CDD, LEVIR-CD, and DSIFN and compare the results with four benchmark methods selected in the literature. A visualization comparison of four datasets is shown in Figs. 6–9. Different colors are used in the figure to represent the detection results: TP (White), TN (black), FP (red), and FN (green).

In the seasonal variation datasets CDD, our model achieves better results than other models and can better adapt to the impact caused by seasonal variation. We select buildings and road changes for analysis. In the comparison of building change results in Fig. 6, the results of our model can extract changing buildings completely even in extreme weather, and changing buildings have the fewest false detection and missed detection. In addition, our model can keep the internal compactness of objects and the integrity of boundaries regardless of large or small changes. In contrast to our model, both U-net++_MSOF and DDCNN damage the internal structure of the large changing area, and the boundary is larger than the real labeled boundary. This is because bitemporal images are concatenated as an input of the network, and image difference discrimination starts from the beginning of the network. Although skip connections concatenate the intermediate feature images, it is difficult to provide the deep features of the original image to help the image reconstruction. IFN can extract changing areas, but there are too many false detections, and the boundary of complex buildings is the most damaged. BiT can maintain the boundary and internal integrity of building changes, but it cannot detect some fine-grained changes, which is due to the loss of details in the changing area caused by the direct up-sampling of the BiT model to the full resolution. For road changes, only our model can detect the changes in narrow roads. For example, U-net++_MSOF, IFN, and BiT models all have serious losses of changed road structures. In addition, the roads detected by DDCNN are larger than the original roads, causing false detection. The change map obtained by our model is closest to the real road change results, and the changing area is complete and accurate, which well suppresses the pseudo-changes.

In DSIFN, the changing dataset of different cities, none of the models can completely maintain the boundary of the changing area, mainly caused by the large image difference between different cities. In Fig. 7, our model has the least amount of missed and false detection in the changing area compared with other models. This is because the transformer

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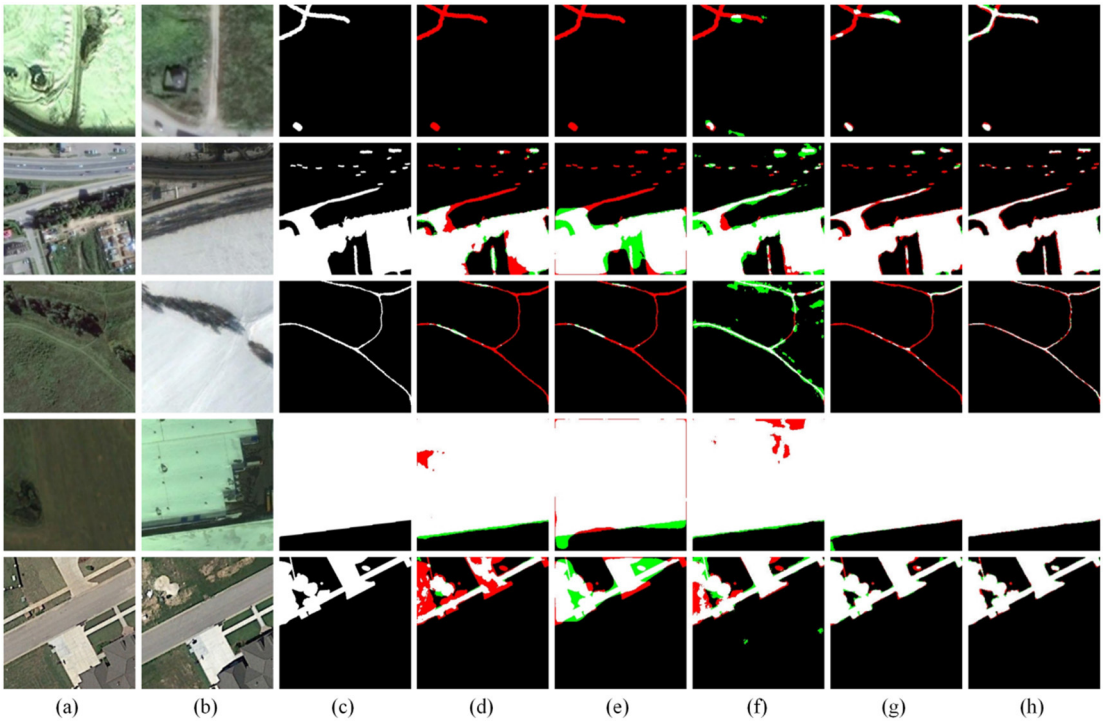


Fig. 6. Visualization results of different methods for CDD datasets. (a) Image T1. (b) Image T2. (c) Ground truth. (d) U-net++_MSOF. (e) IFN. (f) DDCNN. (g) BiT. (h) Ours. In the figure, white, black, red, and green represent TP, TN, FP, and FN, respectively.

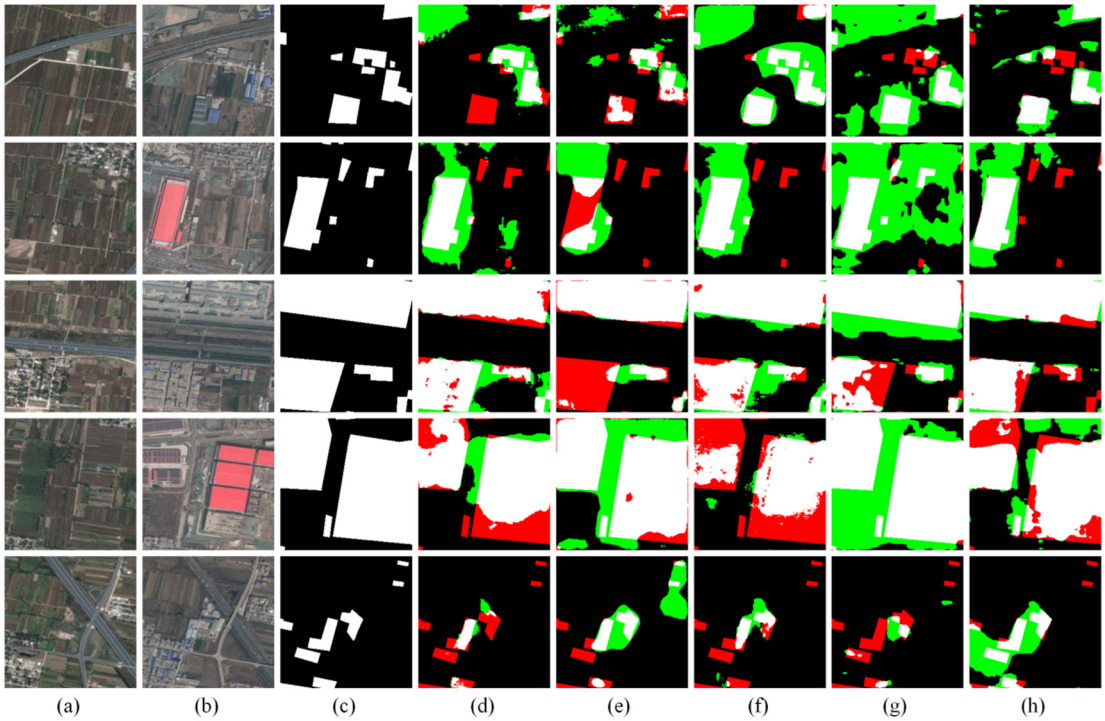


Fig. 7. Visualization results of different methods for DSIFN datasets. (a) Image T1. (b) Image T2. (c) Ground truth. (d) U-net++_MSOF. (e) IFN. (f) DDCNN. (g) BiT. (h) Ours. In the figure, white, black, red, and green represent TP, TN, FP, and FN, respectively.

module is used to learn the context, which eliminates a lot of pseudo-changes and more accurately expresses the real semantic changes. However, many changes cannot be completely detected in the comparison methods U-net++_MSOF, IFN, and DDCNN, and the BiT is typically sensitive to noise, which leads to pseudo-changes in detection results.

In building change datasets WHU-CD and LEVIR-CD, most models can detect large building changes except DDCNN.

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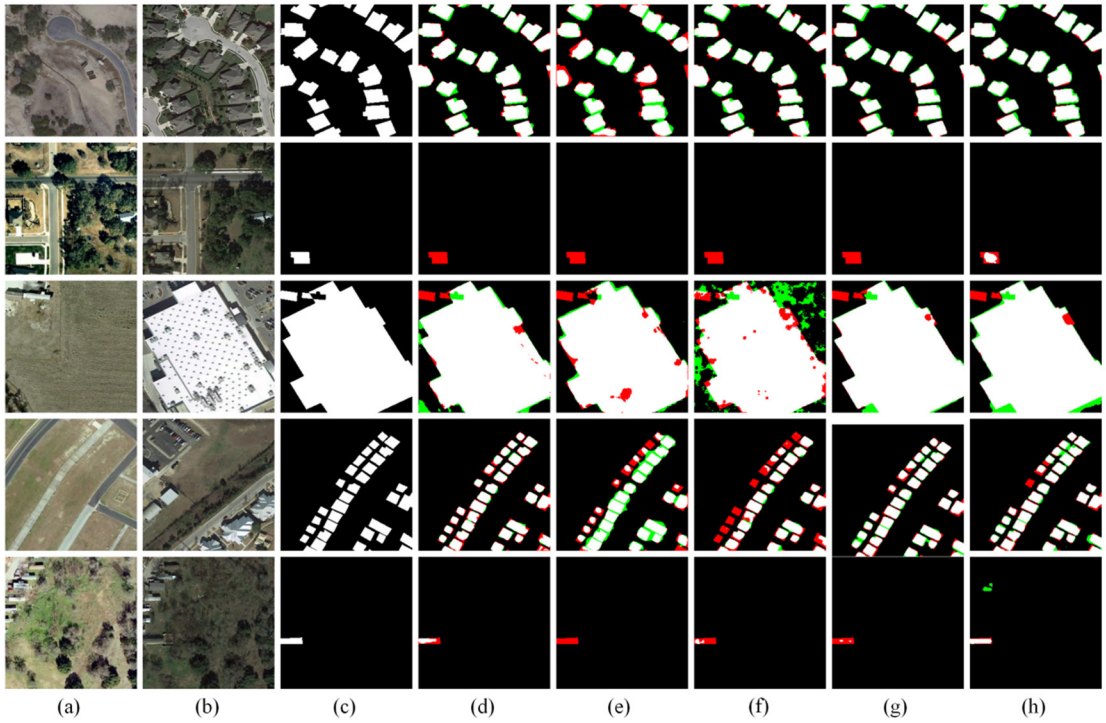


Fig. 8. Visualization results of different methods for LEVIR-CD datasets. (a) Image T1. (b) Image T2. (c) Ground truth. (d) U-net++_MSOF. (e) IFN. (f) DDCNN. (g) BiT. (h) Ours. In the figure, white, black, red, and green represent TP, TN, FP, and FN, respectively.

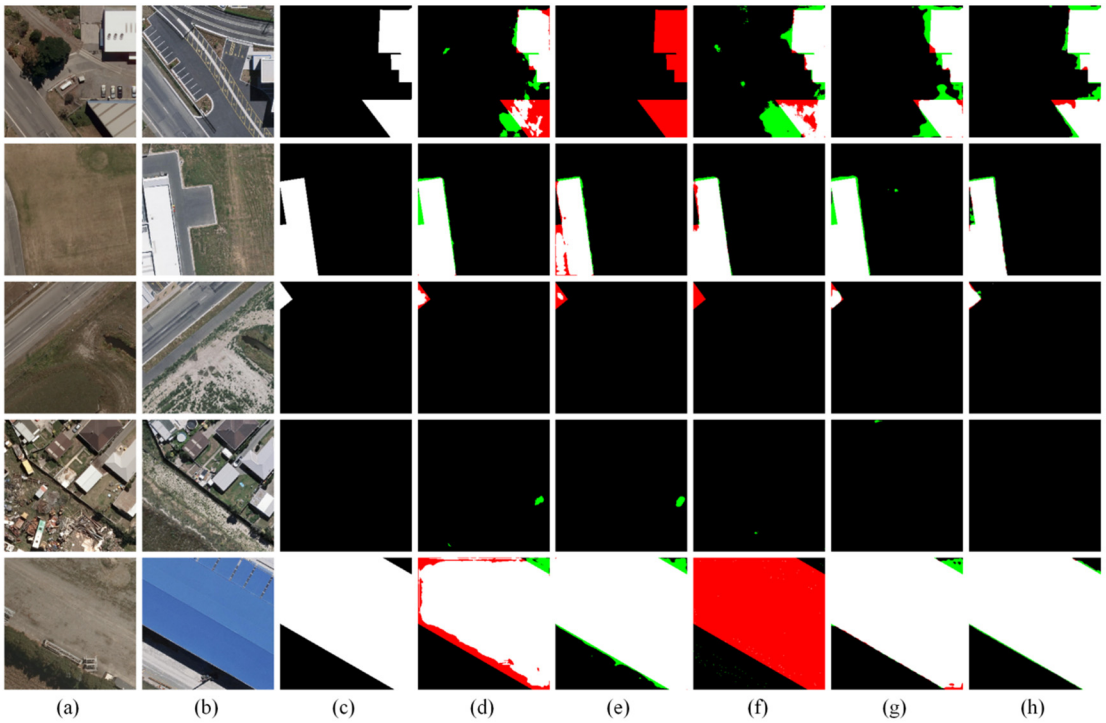


Fig. 9. Visualization results of different methods for WHU-CD datasets. (a) Image T1. (b) Image T2. (c) Ground truth. (d) U-net++_MSOF. (e) IFN. (f) DDCNN. (g) BiT. (h) Ours. In the figure, white, black, red, and green represent TP, TN, FP, and FN, respectively.

Among them, U-net++_MSOF generates the missing edge, IFN suffers from false alarms and missing detection of complex buildings, and BiT and our model can detect changes accurately and weaken the influence of noise. For small building changes, only our model can almost distinguish all small changed objects. This is because the U-net++_MSOF, IFN, and DDCNN models are limited by receptive field, while the BiT model is limited by the loss of low-level

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TABLE I
QUANTITATIVE COMPARISON RESULTS OF CDD DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
U-net++_MSOF	86.68	76.53	81.29	68.48
IFN	90.56	70.18	79.08	65.40
DDCNN	89.18	82.14	85.45	74.59
BiT	96.19	93.99	95.07	90.61
Ours	**96.93**	**97.42**	**97.17**	**94.50**

TABLE II
QUANTITATIVE COMPARISON RESULTS OF DSIFN DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
U-net++_MSOF	74.93	65.74	55.33	54.42
IFN	75.22	60.93	53.02	50.97
DDCNN	**79.27**	67.65	58.45	**59.30**
BiT	78.16	58.56	60.00	52.76
Ours	71.55	**69.42**	**66.62**	57.95

TABLE III
QUANTITATIVE COMPARISON RESULTS OF LEVIR-CD DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
U-net++_MSOF	90.33	81.82	85.86	75.24
IFN	90.61	55.73	69.86	53.69
DDCNN	88.52	81.39	84.81	73.62
BiT	92.04	87.96	89.96	81.75
Ours	**92.43**	**89.82**	**91.11**	**83.67**

TABLE IV
QUANTITATIVE COMPARISON RESULTS OF WHU-CD DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
U-net++_MSOF	91.39	80.06	85.35	74.44
IFN	90.27	75.53	82.24	69.84
DDCNN	88.52	81.39	84.81	73.62
BiT	90.65	88.30	88.30	76.86
Ours	**93.59**	**89.60**	**93.59**	**84.42**

details in direct up-sampling, resulting in the appearance of pseudo-changes and the loss of boundaries. As shown in Fig. 8, the changes in small white buildings in LEVIR-CD are not detected by other models except ours.

Tables I–IV compare the quantitative verification results of the proposed model and the comparison models. In the CDD, WHU-CD, and LEVIR-CD datasets, the proposed model reaches the optimal level in all four metrics. In the CDD dataset, the precision, recall, F1 score, and IoU are 96.93%, 97.42%, 97.17%, and 94.50%, respectively, which exceed the benchmark method by 10.25%, 20.89%, 15.88%, and 26.02% and outperform the model (BiT) of the same type by 0.74%, 3.43%, 2.10%, and 3.89%. In the DSIFN dataset, while the detection accuracy of all methods is not high, our model can detect more changing pixels, which makes a good tradeoff between precision and recall rate. In the LEVIR-CD and WHU-CD building datasets, our model outperforms the

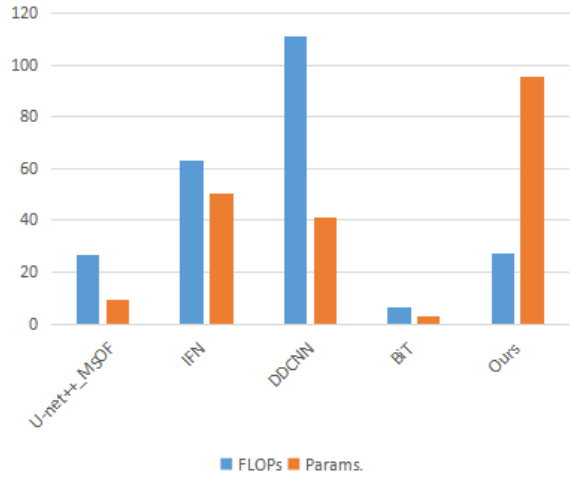


Fig. 10. Params and FLOPs comparison of different methods.

benchmark methods by 2.10%, 8.00%, 5.25%, 8.43%, and 2.20%, 9.54%, 8.24%, 9.98%, respectively. In the quantitative comparison of various datasets, our model shows the best performance, with clear boundaries, high internal compactness, and fewer missed and false detections.

2) Parameter Comparison: This article compares the number of parameters and computational complexity of different models. Fig. 10 shows the number of parameters (Params.) and floating-point operations per second (FLOP) of each model. Compared with other models, the computational complexity of our model is lower, and model parameters are more, which is caused by using ViT as the pretraining model. However, compared with the improved accuracy of the model, the impact of the increased parameters is negligible.

3) Learning Curve Comparison: Fig. 11 compares training visualization results between our model and the BiT model. The figure shows the mean F1-score of the training set and the validation set of 200 epochs. From the comparison of training datasets, our model makes a faster convergence speed and can achieve the optimal training results more efficiently with higher training accuracy. From the comparison of validation datasets, our model is more stable and has good generalization ability, but the BiT model oscillates greatly with the increase of epoch, resulting in poor robustness. This also proves the great learning and generalization ability of the transformer and UNet network to CD.

C. Ablation Study

1) Fusion Strategy Experiment: In order to verify the best experiment of the transformer in the UNet network, we design three fusion strategies for comparative verification, including EF, late-fusion-conc, and late-fusion-diff. In EF, images obtained at different times are concatenated into an input image. Late-fusion-conc and late-fusion-diff adopt two skip connections methods, respectively, after feature extraction, namely concatenation and difference. According to Fig. 12 and Table V, the F1-score of EF strategy is the worst, 1%–8% lower than that of late fusion. The EF results in a loss of detail and false alarms, such as street information missing in

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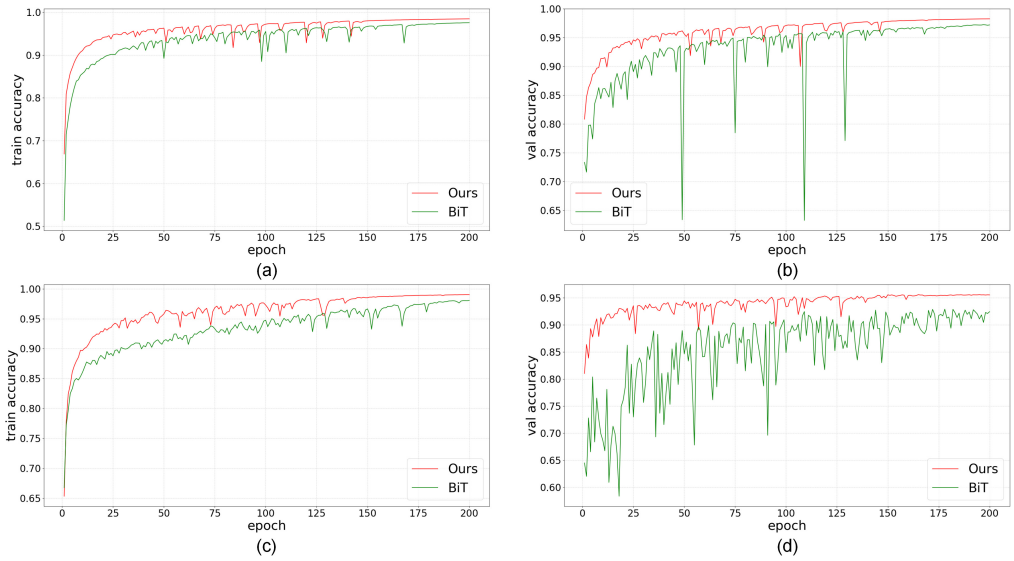


Fig. 11. Learning curve visualization comparison. (a) CDD training set the learning curve. (b) CDD verification set the learning curve. (c) WHU-CD training set t learning curve. (d) WHU-CD verification set the learning curve.

TABLE V
QUANTITATIVE COMPARISON RESULTS OF FUSION STRATEGY

Fusion strategy	CDD P / R / F1 / IoU (%)	LEVIR-CD P / R / F1 / IoU (%)	WHU-CD P / R / F1 / IoU (%)
Early fusion	96.00/95.52/95.76/91.86	90.93/87.29/89.07/80.29	93.47/78.58/85.38/74.49
Late fusion-diff	96.66/97.07/96.86/93.91	91.14/89.65/90.39/82.46	**93.68**/87.92/90.71/83.00
Late fusion-conc	**96.93**/**97.42**/**97.17**/**94.50**	**92.43**/**89.82**/**91.11**/**83.67**	93.59/**89.60**/**93.59**/**84.42**

TABLE VI
QUANTITATIVE COMPARISON RESULTS OF MODULE VERIFICATION

Transformer	DEM	CDD P / R / F1 / IoU (%)	LEVIR-CD P / R / F1 / IoU (%)	WHU-CD P / R / F1 / IoU (%)
×	×	96.72/97.18/96.95/94.09	90.18/90.15/90.46/82.49	**95.26**/87.74/91.35/84.08
×	√	96.79/97.09/96.96/94.11	90.17/**90.19**/90.55/82.53	94.87/88.00/92.15/84.06
√	×	96.90/97.33/96.99/94.35	90.36/89.89/90.72/82.80	93.29/88.76/93.32/84.16
√	√	**96.93**/**97.42**/**97.17**/**94.50**	**92.43**/89.82/**91.11**/**83.67**	93.59/**89.60**/**93.59**/**84.42**

CDD datasets, roof misdetection in LEVIR-CD datasets, and new building missing in WHU-CD datasets. This is mainly because the EF cannot present the deep features of a single original image, which is not conducive to highlighting the changed information. Late-fusion-diff is slightly less accurate than late-fusion-conc, and it generates small buildings missing detection, which is due to subtracting part of effective information from the difference map. Therefore, this article adopts the late-fusion-conc strategy.

2) Module Verification Experiment: We design ablation experiments for each module to verify the effectiveness of each module, including the transformer module and the difference enhancement (DE) module, as shown in Table VI. Experiments show that the transformer and DEM are effective and robust to CD algorithms. This is mainly because the transformer module can aggregate global information, accurately locate changing areas, and be more robust to texture, shape, and size features. The existence of the DE module can improve the accuracy and make the model converge quickly in the early stage.

3) Parameter Verification Experiment: We use CDD datasets for parameter verification experiments, as shown in Table VII.

a) Effect of position embedding: This experiment analyzes the necessity of position embedding for CD. Position embedding introduces spatial location information, which is effective for positioning change location. The experimental results show that the accuracy is improved after the position embedding is added.

b) Effect of patch size: We study the impact of patch size on CD accuracy. Here, we design patch sizes as 8, 16, and 32, respectively. The experimental results show that the smaller the patch size is, the higher the detection effect is. This is because the transformer encodes more complex dependencies

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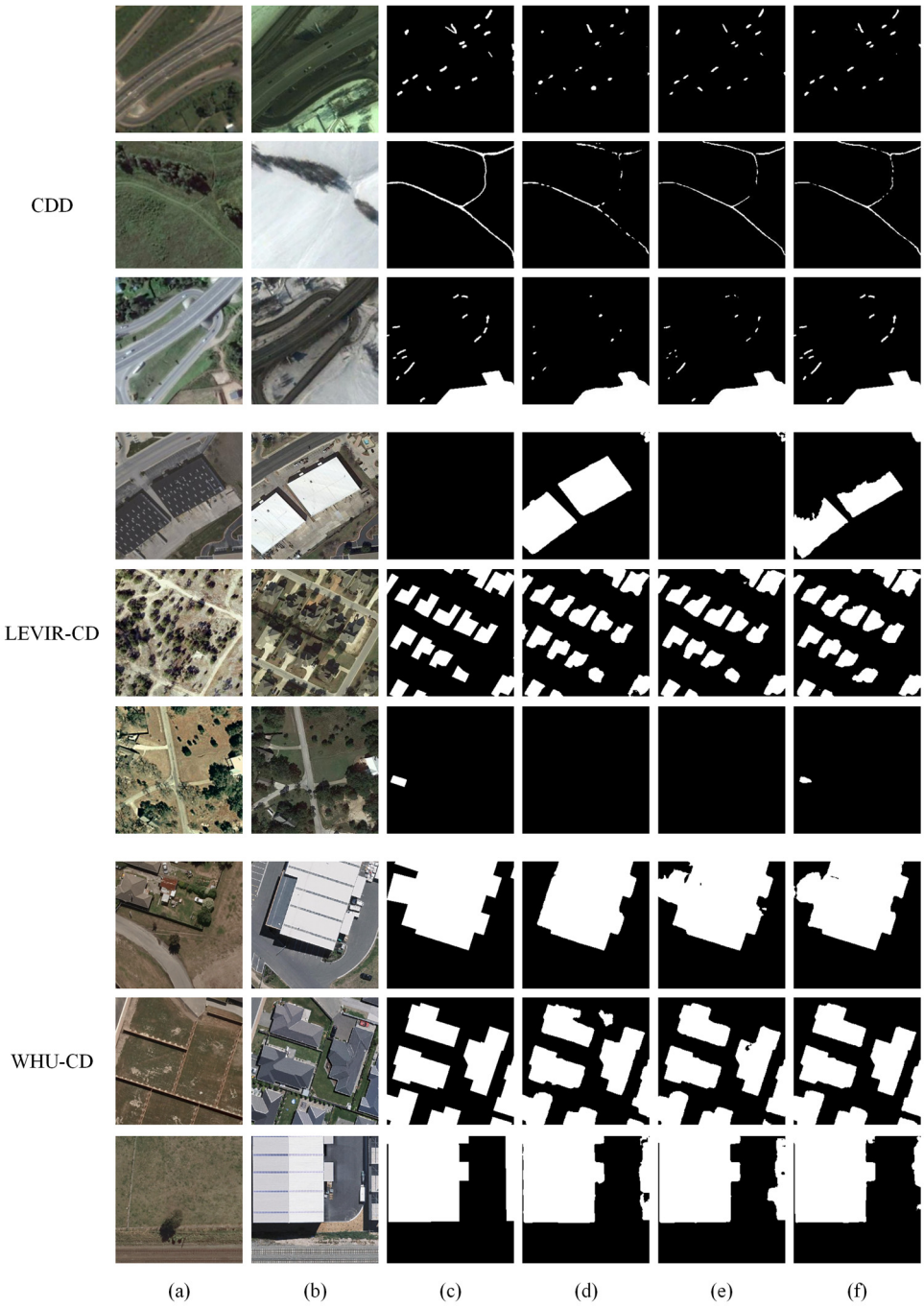


Fig. 12. Fusion strategy comparison. (a) Image T1. (b) Image T2. (c) Ground truth. (d) EF. (e) Late fusion-diff. (f) Late fusion-conc.

between each element for longer input sequences. However, the increase in patch size affects the training efficiency. Therefore, we choose 16 as the most appropriate patch size.

c) Effect of transformer depth: The increase of transformer depth improves the CD accuracy slightly, but it brings a rapid increase in parameters and FLOPs. Here, we set transformer depth to 12 to balance efficiency and accuracy.

d) Effect of the number of skip connections: We study the impact of skip connections on CD results, where 0 means no skip connections, 1 means only 1/2 skip connections, 2 means 1/2 and 1/4 skip connections, and 3 represents the model in this article. Experimental results show that adding more skip connections can bring better performance and locate the change details and texture features more accurately. Skip connections are beneficial to reduce the missing and false edge details and improve the accuracy of small change images.

D. Additional Analysis

1) Feature Visualization: We use the Grad-Cam++ [58] method to visualize the feature map before and after adding the transformer. Red indicates higher attention values and blue

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TABLE VII
QUANTITATIVE COMPARISON RESULTS OF PARAMETER VERIFICATION

Parameter	Index	Params(M)	FLOPs(G)	P(%)	R(%)	F1(%)	IoU(%)
position embedding	×	95.49	27.23	96.85	97.34	97.05	94.34
	√	95.49	27.23	96.93	97.42	97.17	94.50
Patch Size	8	95.49	27.23	96.75	97.52	97.19	94.53
	16	95.49	27.23	96.93	97.42	97.17	94.50
	32	95.49	27.23	97.00	97.25	97.12	94.40
Transformer depth	6	52.98	18.91	97.04	97.37	97.01	94.47
	12	95.49	27.23	96.93	97.42	97.17	94.50
	18	138.00	35.55	96.67	97.73	97.19	94.54
the number of skip connections	0	92.47	22.61	96.61	95.84	96.23	92.73
	1	94.83	24.46	96.96	97.16	97.06	94.29
	2	95.42	26.31	96.95	97.21	97.11	94.31
	3	95.49	27.23	96.93	97.42	97.17	94.50

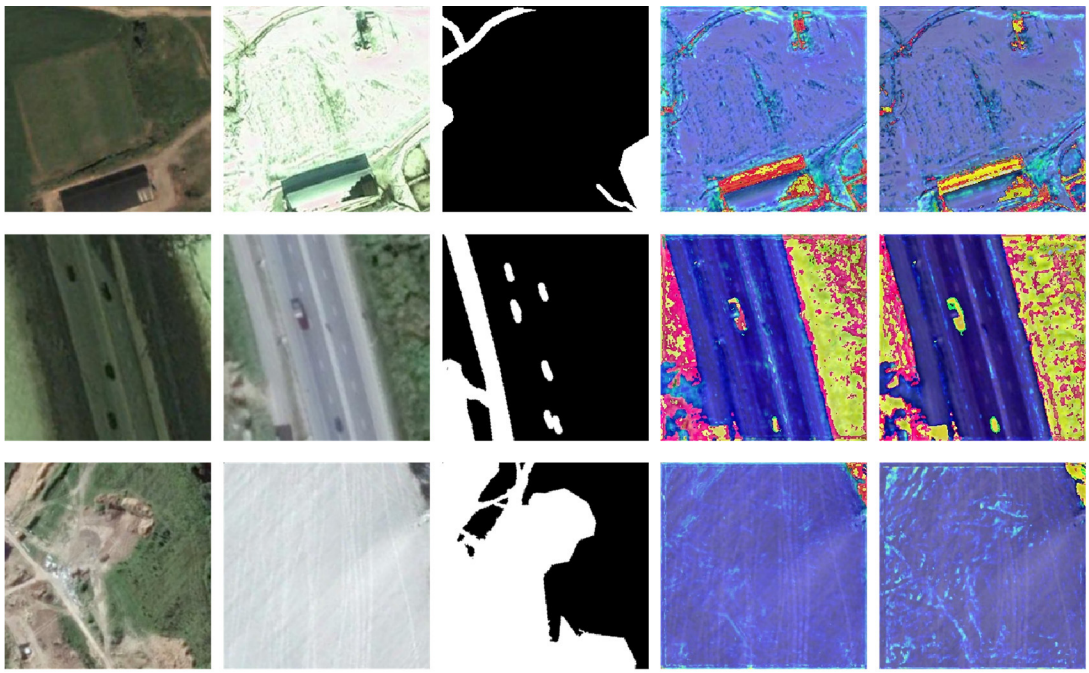


Fig. 13. Feature visualization. (a) Image T1. (b) Image T2. (c) Ground truth. (d) No Transformer. (e) Adding Transformer. The intensity of features in the figure is related to brightness. Red indicates higher attention values and blue indicates lower attention values. The brighter the color, the stronger the feature.

indicates lower attention values. Fig. 13 shows that our model can highlight different semantic features, such as buildings, cars, and farmlands. After the transformer, the changing area of the feature map has higher energy, and the area with the largest weight is mainly located at the edge of the changing area. The transformer can better locate the changing area, enhance the expression ability of the changing area edge, and improve the determination of the changing object edge and small object detection.

From the comparison before and after the addition of the transformer, some background noise is concerned when there is no transformer, and adding the transformer can better suppress background noise, such as vegetation according to the seasonal change. That is, the transformer can filter out redundant context information in low-level features and enhance high-level semantic features.

TABLE VIII
QUANTITATIVE COMPARISON RESULTS OF LR EXPERIMENT

Method	P(%)	R(%)	F1(%)	IoU(%)
Ours	**96.93**	**97.42**	**97.17**	**94.50**
4-times	96.38	96.19	96.28	92.83
8-times	94.44	89.42	91.86	84.94
BiT	96.19	93.99	95.07	90.61

2) Reduced Resolution Experiment: This experiment is designed to simulate bitemporal images with

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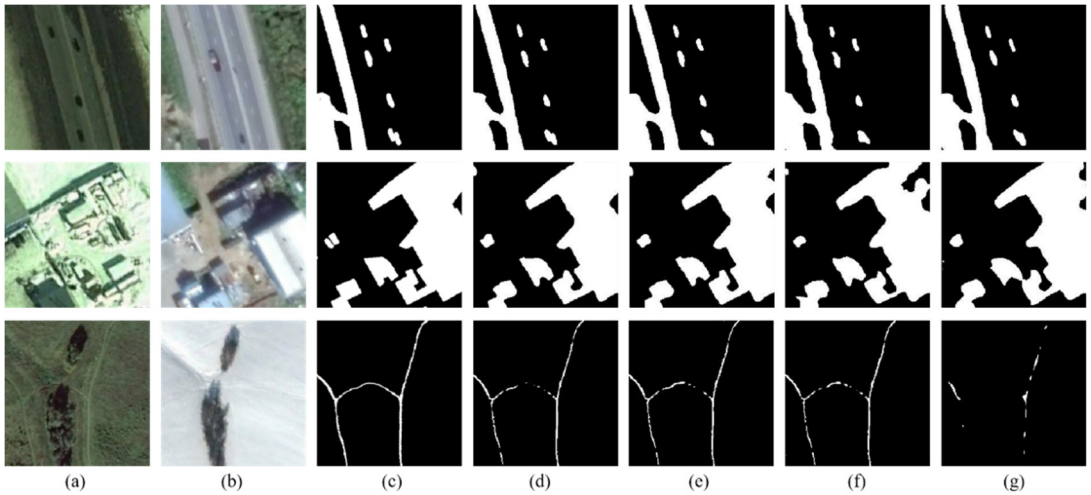


Fig. 14. Visualization results of the LR experiment. (a) Image T1. (b) Image T2. (c) Ground truth. (d) Ours. (e) 4-times-resolution-difference. (f) 8-times-resolution-difference. (g) BiT.

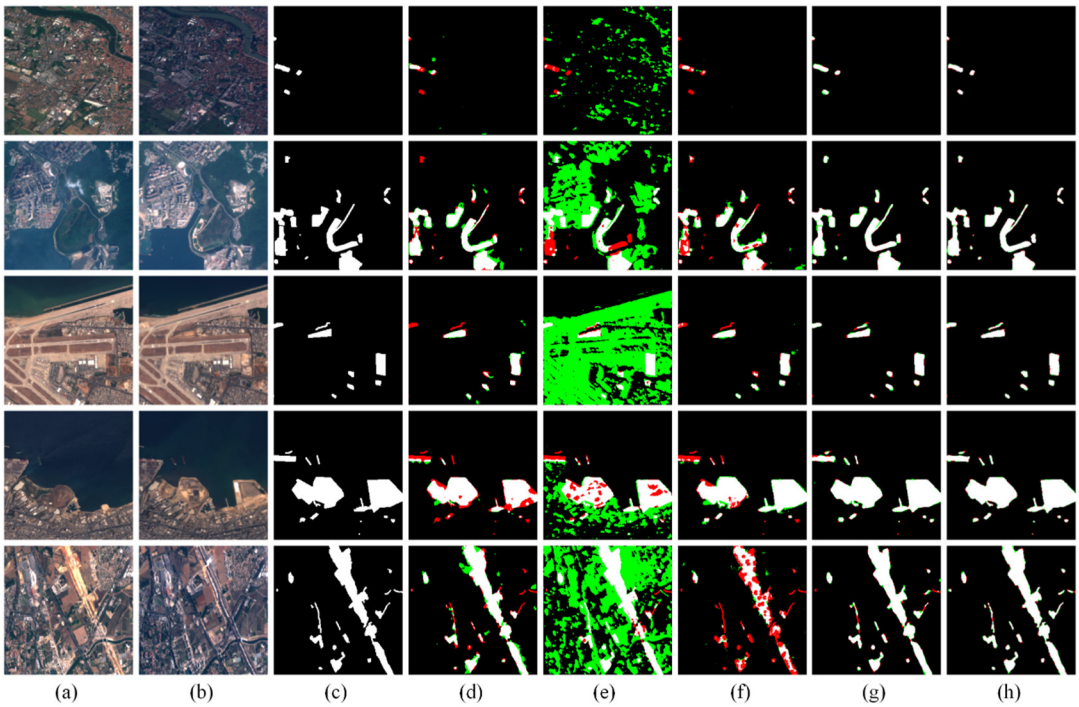


Fig. 15. Visualization results of different methods for OSCD datasets. (a) Image T1. (b) Image T2. (c) Ground truth. (d) U-net++_MSOF. (e) IFN. (f) DDCNN. (g) BiT. (h) Ours.

4-times-resolution-difference and 8-times-resolution-difference. We downsampled the images for the first time in CDD datasets 4/8 times to obtain reduced resolution (LR) images. These LR images are interpolated to obtain the original image size, which is then used as input. Based on this, we verify the performance of the proposed model under large spatial differences and prove its robustness and universality. The experimental results are shown in Fig. 14 and Table VIII. The experimental results show that 4-times-resolution-difference and 8-times-resolution-difference are 0.55%, 1.23%, 0.89%, 1.67% and 2.49%, 8.00%, 5.31%, 9.56% lower than no-resolution-difference in P, R, F1, and IoU, respectively. After reducing the resolution, the CD result generates pixel loss at the edge, but the changing area can be completely detected. The detection accuracy is better than that of the BiT model in narrow roads, which proves its generalization ability.

3) Real Medium Satellite Datasets Validation Experiment: We use OSCD, a Sentinel-2 dataset, to verify the accuracy and robustness of medium-resolution data CD. The experimental results are shown in Fig. 15 and Table IX. We draw the following conclusions that the accuracy of our model is the best in all metrics, surpassing the benchmark model by 2.3%, 12.68%, 7.86%, and 12.85%. In addition, the proposed model

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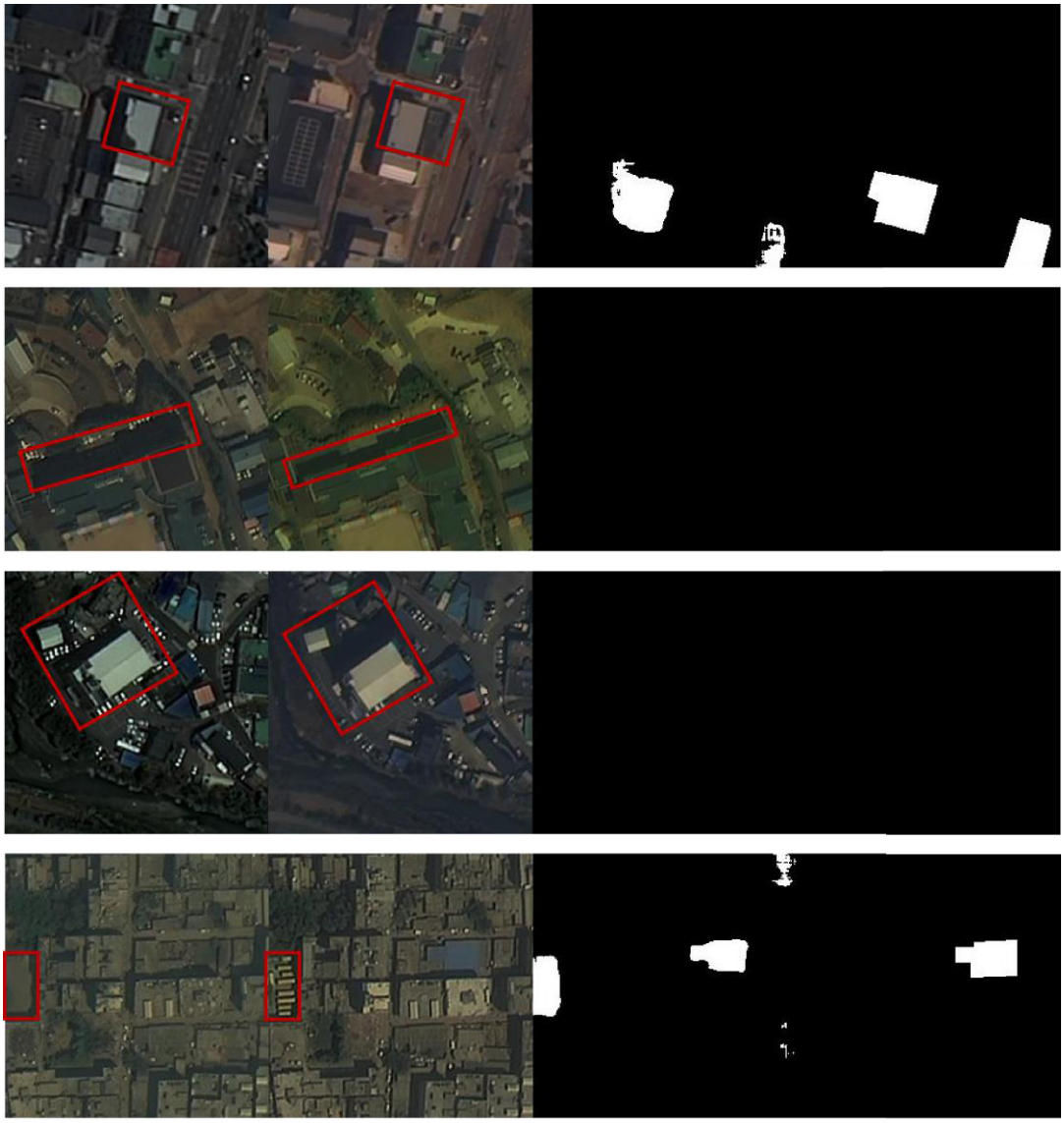


Fig. 16. Visualization results of multilooking satellite datasets. (a) Image T1. (b) Image T2. (c) Ours. (d) Ground truth. In the figure, we mark the effect of the change in perspective in red.

TABLE IX
QUANTITATIVE COMPARISON RESULTS OF OSCD DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
U-net++_MSOF	91.39	80.06	85.35	74.44
IFN	90.27	75.53	82.24	69.84
DDCNN	88.52	81.39	84.81	73.62
BiT	90.65	88.30	88.30	76.86
Ours	**93.69**	**92.74**	**93.21**	**87.29**

TABLE X
QUANTITATIVE COMPARISON RESULTS OF S2 LOOKING DATASETS

Method	P(%)	R(%)	F1(%)	IoU(%)
BiT	72.65	53.80	61.85	50.16
Ours	**76.41**	**59.70**	**67.03**	**54.41**

outperforms the BiT model by 3.04%, 4.44%, 4.91%, and 10.43%. The IFN model generates many false alarms, which is due to the low resolution of Sentinel-2 and the complexity of urban areas. The U-net++_MSOF and DDCNN model suffer from partially missed detection, and the internal structure is damaged. The BiT model loses the details of fine changes. Our model returns the changing area with complete boundary and high internal compactness, which is closest to the label sample.

4) Multilooking Satellite Datasets Validation Experiment: We use the S2Looking datasets to verify the CD of the multilooking satellite image, as shown in Fig. 16. Since the datasets contain large-scale satellite side-view images captured

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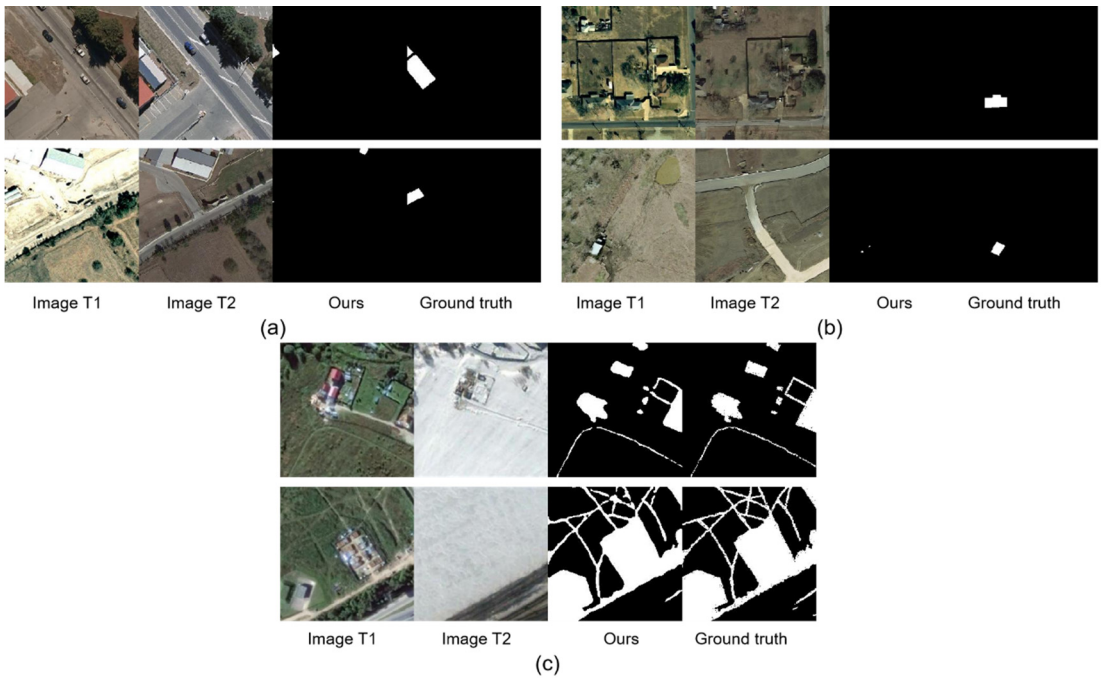


Fig. 17. Some weaknesses of our model. (a) Errors caused by similar color and brightness. (b) Errors caused by building decline. (c) Errors caused by narrow roads.

at different angles from the lowest point, it brings difficulties to CD. The figure shows the changes in the shape, position, and shadow of buildings in the images before and after due to the changes in the perspective of satellite shooting. According to Table X, compared with the BiT model, our model improves by 3.76%, 5.90%, 5.18%, and 4.25 in P, R, F1, and IoU, respectively. Our model is better able to identify changing areas, which can mitigate pseudo-changes caused by building shapes from different perspectives. In addition, the model is more robust to building shadows and seasonal changes. Finally, our model is better at detecting small changes in the established boundaries between bi-temporal images.

5) Some weaknesses: During the experiment, we also find some possible problems. In Fig. 17(a), our model cannot detect some building changes with similar color and brightness. At the same time, for some objects that have great changes due to season, our model produces false detection. In Fig. 17(b), we can see that the detection accuracy of our model is poor for building decline. In Fig. 17(c), some narrow roads and complex roads we detected may result in discontinuity and merging of roads.

V. Conclusion

In this article, we propose a hybrid transformer network for remote-sensing image change detection (TransUNetCD). Our model introduces the transformer into the encoder–decoder to model the context and obtain rich global context information. Combined with the UNet decoder, our model enhances the low-level details by recovering local spatial information and realizes accurate localization, improving the edge expression ability of the changing area. On this basis, the DEM is introduced to weight the change map to realize the fast convergence and improve the learning ability of the network.

The experimental results show that the proposed model has good robustness and generalization and outperforms all benchmark comparison methods in multiple datasets, which proves the effectiveness of the transformer in the change detection network. In the seasonal variation datasets, compared with other methods, our model can better identify the pseudo-seasonal variation. In the building datasets, our model can better preserve the edges and interior structures of buildings compared with other methods. At the same time, in the medium-resolution satellite image datasets, our model also maintains good detection accuracy. Therefore, our model can adapt to different application scenarios, such as urban planning and land use/cover. Compared with models without the transformer, the improved accuracy of our model demonstrates the elimination of redundant information and the full modeling of features. Compared with models with the transformer, our model has the ability to restore changing areas' details and maintain changing areas' edges. At the same time, the learning curves of our model in the training set and verification set prove the generalization and robustness. In addition, our model also shows good detection ability in the change detection of images with different resolutions, which can be used for disaster monitoring and other time-sensitive tasks.

Future work will study the adaptability of the model in heterogeneous images, such as optical satellite, synthetic aperture radar (SAR) satellite, and unmanned aerial vehicle (UAV) images. At the same time, the weakening of buildings, the pseudo-change of color approximation, and other problems need further study.

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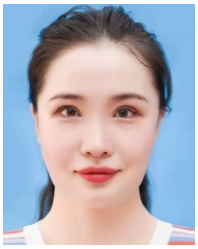
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